

CSA0704 | CISCO ENTERPRISE NETWORK SIMULATOR

Python GUI Front-End | Cisco-Style Network Verification

TOPOLOGY

IP / VLSM

ROUTING

PING

TCP / UDP

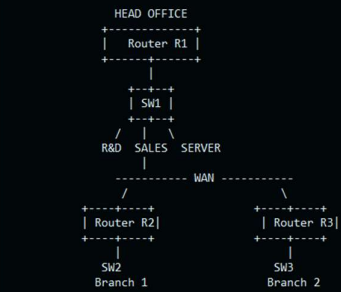
HTTP / DNS

TRAFFIC

LINK FAILURE

RUN ALL

CISCO-STYLE COMMAND / RESULT OUTPUT



Design: Hybrid enterprise topology
Devices: 3 Routers, 3 Switches, PCs and 1 Server

★ TOPOLOGY COMPLETED

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R1# show ip interface brief

Interface	IP Address	Status
G0/0	192.168.10.1	up
G0/1	10.0.0.1	up
G0/2	10.0.0.5	up

VLSM SUBNETS

R&D	192.168.10.0/28
SALES	192.168.10.16/28
HR	192.168.10.32/29
FINANCE	192.168.10.40/29
BRANCH 1	192.168.10.48/29
BRANCH 2	192.168.10.56/29

RESULT: ADDRESSING VERIFIED

★ IP / VLSM COMPLETED

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```
R1# show ip route
C 192.168.10.0/28   directly connected
C 192.168.10.16/28  directly connected
C 192.168.10.32/29  directly connected
S 192.168.10.48/29  via 10.0.0.2
S 192.168.10.56/29  via 10.0.0.6

R2# show ip route
S 192.168.10.0/28   via 10.0.0.1
S 192.168.10.16/28  via 10.0.0.1

R3# show ip route
S 192.168.10.0/28   via 10.0.0.5

RESULT: ROUTING TABLE VERIFIED
```

• ROUTING COMPLETED

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```
PC-R&D> ping 192.168.10.50

Reply from 192.168.10.50: bytes=32 time<1ms
Reply from 192.168.10.50: bytes=32 time<1ms
Reply from 192.168.10.50: bytes=32 time<1ms
Reply from 192.168.10.50: bytes=32 time<1ms

Sent = 4
Received = 4
Lost = 0 (0% loss)

RESULT: END-TO-END CONNECTIVITY SUCCESSFUL
```

• PING TEST COMPLETED

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PROTOCOL TEST

TCP

Connection : Established
Delivery : Reliable
Status : PASS

UDP

Connection : Connectionless
Overhead : Low
Status : PASS

TCP -> reliable and ordered delivery
UDP -> faster, connectionless delivery

RESULT: TCP / UDP ANALYSIS COMPLETED

• TCP / UDP COMPLETED

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SERVER1

IP Address : 192.168.10.12

HTTP SERVICE

Status : ON
Request: http://192.168.10.12
Result : Web response received

DNS SERVICE

Status : ON
www.enterprise.local -> 192.168.10.12
Result : Name resolved

RESULT: HTTP / DNS VERIFIED

• HTTP / DNS COMPLETED

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PERFORMANCE TEST

Condition	Latency	Throughput	Packet Loss
Normal	18 ms	92 Mbps	1%
High Traffic	35 ms	78 Mbps	4%
Congestion	58 ms	61 Mbps	9%

Observation:

As traffic increases, latency and packet loss increase, while available throughput decreases.

RESULT: PERFORMANCE ANALYSIS COMPLETED

• TRAFFIC COMPLETED

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LINK FAILURE TEST

BEFORE FAILURE

PC -> Branch PC : SUCCESS

LINK DOWN

R1 G0/1 ----- X ----- R2 G0/1

DURING FAILURE

PC -> Branch PC : REQUEST TIMED OUT

RECOVERY

Link restored

PC -> Branch PC : SUCCESS

RESULT: FAILURE AND RECOVERY VERIFIED

• FAILURE COMPLETED

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- TRAFFIC
- LINK FAILURE
- RUN ALL

CISCO-STYLE COMMAND / RESULT OUTPUT

```
=====
ENTERPRISE NETWORK - FINAL VERIFICATION
=====

IPv4 / VLSM      : PASS
Routing          : PASS
Ping Connectivity : PASS
TCP              : PASS
UDP              : PASS
HTTP             : PASS
DNS              : PASS
Traffic Analysis : TESTED
Link Failure / Recovery : TESTED

Normal   : 18 ms | 92 Mbps | 1% loss
High     : 35 ms | 78 Mbps | 4% loss
Congested : 58 ms | 61 Mbps | 9% loss

FINAL STATUS: NETWORK TESTING COMPLETED
=====
```

• ALL TESTS COMPLETED